
Snippet 1:

```
public class Main {  
    public void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

**Error: main method not found in class Main, please define the main method as:
public static void main(String[] args)**

Corrected code

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

Snippet 2:

```
public class Main {  
    static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

**Error: main method not found in class Main, please define the main method as:
public static void main(String[] args)**

Corrected Code

```
public class Main {  
    public static void main(String[] args)  
    {  
        System.out.println("Hello, World!");  
    }  
}
```

Snippet 3:

```
public class Main {  
    public static int main(String[] args) {  
        System.out.println("Hello, World!");  
        return 0;  
    }  
}
```

**Error: main method not found in class Main, please define the main method as:
public static void main(String[] args)**

Corrected Code

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

Snippet 4:

```
public class Main {  
    public static void main() {  
        System.out.println("Hello, World!");  
    }  
}
```

**Error: main method not found in class Main, please define the main method as:
public static void main(String[] args)**

Corrected Code

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

Snippet 5:

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("Main method with String[] args");  
    }  
}
```

```

    }
    public static void main(int[] args) {
        System.out.println("Overloaded main method with int[] args");
    }
}

```

Error: main method not found in class Main, please define the main method as: public static void main(String[] args)

Corrected Code

```

public class Main {
    public static void main(String[] args) {
        System.out.println("Main method with String[] args");
    }
}

```

// Overloading main method for different parameter types, but this won't be called by the JVM.

```

    public static void main(int[] args) {
        System.out.println("Overloaded main method with int[] args");
    }
}

```

Snippet 6:

```

public class Main {
    public static void main(String[] args) {
        int x = y + 10;
        System.out.println(x);
    }
}

```

error: cannot find symbol: variable y

Corrected code

```

public class Main {
    public static void main(String[] args) {
        int y = 5; // Declare and initialize y
        int x = y + 10;
        System.out.println(x);
    }
}

```

```
}  
}
```

Snippet 7:

```
public class Main {  
    public static void main(String[] args) {  
        int x = "Hello";  
        System.out.println(x);  
    }  
}
```

error: incompatible types: String cannot be converted to int

Corrected code

```
public class Main {  
  
    public static void main(String[] args) {  
  
        String x = "Hello"; // Declare x as a String  
  
        System.out.println(x);  
  
    }  
}
```

Snippet 8:

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!")  
    }  
}
```

Error: reached end of file while parsing

Corrected Code

```
public class Main {  
    public static void main(String[] args) {
```

```
        System.out.println("Hello, World!"); // Added missing closing parenthesis and
    }
}
```

Snippet 9:

```
public class Main {
    public static void main(String[] args) {
        int class = 10;
        System.out.println(class);
    }
}
```

Error: class is a reserved keyword in Java, and you cannot use it as an identifier for variables or methods.

Corrected Code

```
public class Main {
    public static void main(String[] args) {
        int classVariable = 10; // Changed 'class' to 'classVariable'
        System.out.println(classVariable);
    }
}
```

Snippet 10:

```
public class Main {
    public void display() {
        System.out.println("No parameters");
    }

    public void display(int num) {
        System.out.println("With parameter: " + num);
    }

    public static void main(String[] args) {
        Main obj = new Main();
        obj.display(); // Calls display() with no parameters
        obj.display(5); // Calls display(int num)
    }
}
```

// Method Overloading is allowed in Java.

Corrected Code

```
public class Main {  
    public void display() {  
        System.out.println("No parameters");  
    }  
  
    public void display(int num) {  
        System.out.println("With parameter: " + num);  
    }  
  
    public static void main(String[] args) {  
        Main obj = new Main();  
        obj.display(); // Calls display() with no parameters  
        obj.display(5); // Calls display(int num)  
    }  
}
```

Snippet 11:

```
public class Main {  
    public static void main(String[] args) {  
        int[] arr = {1, 2, 3};  
        System.out.println(arr[5]);  
    }  
}
```

Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: Index 5 out of bounds for length 3

Corrected Code

```
public class Main {  
    public static void main(String[] args) {  
        int[] arr = {1, 2, 3};  
        System.out.println(arr[2]); // Accessing a valid index  
    }  
}
```

Snippet 12:

```
public class Main {  
    public static void main(String[] args) {  
        while (true) {  
            System.out.println("Infinite Loop");  
        }  
    }  
}
```

// The code will run infinitely and print "Infinite Loop" continuously.

Corrected Code

```
public class Main {  
    public static void main(String[] args) {  
        for (int i = 0; i < 10; i++) {  
            System.out.println("Loop " + (i + 1));  
        }  
    }  
}
```

Snippet 13:

```
public class Main {  
    public static void main(String[] args) {  
        String str = null;  
        System.out.println(str.length());  
    }  
}
```

Exception in thread "main" java.lang.NullPointerException.

Corrected Code

```
public class Main {  
    public static void main(String[] args) {  
        String str = "Hello";  
        System.out.println(str.length()); // Now str is not null  
    }  
}
```

Snippet 14:

```
public class Main {  
    public static void main(String[] args) {  
        double num = "Hello";  
        System.out.println(num);  
    }  
}
```

error: incompatible types: String cannot be converted to double

Corrected Code

```
public class Main {  
    public static void main(String[] args) {  
        String str = "Hello";  
        System.out.println(str.length()); // Now str is not null  
    }  
}
```

Snippet 15:

```
public class Main {  
    public static void main(String[] args) {  
        int num1 = 10;  
        double num2 = 5.5;  
        double result = num1 + num2; // Corrected to handle mixed data types  
        System.out.println(result);  
    }  
}
```

error: incompatible types: int cannot be converted to double

Corrected Code

```
public class Main {  
    public static void main(String[] args) {  
        int num1 = 10;  
        double num2 = 5.5;  
        double result = num1 + num2; // Corrected to handle different data types  
        System.out.println(result);  
    }  
}
```

Snippet 16:

```
public class Main {  
    public static void main(String[] args) {  
        int num = 10;  
        double result = num / 4.0; // Use 4.0 to ensure floating-point division  
        System.out.println(result);  
    }  
}
```

// The output will be 2.5

Corrected Code

```
public class Main {  
    public static void main(String[] args) {  
        int num = 10;  
        double result = num / 4.0; // Use 4.0 to ensure floating-point division  
        System.out.println(result);  
    }  
}
```

Snippet 17:

```
public class Main {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 5;  
        int result = a ** b; // Invalid operator  
        System.out.println(result);  
    }  
}
```

error: operator '' is undefined for the type int**

Corrected Code

```
public class Main {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 5;  
        double result = Math.pow(a, b); // Use Math.pow() for exponentiation  
        System.out.println(result);  
    }  
}
```

```
}
```

Snippet 18:

```
public class Main {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 5;  
        int result = a + b * 2; // Operator precedence  
        System.out.println(result);  
    }  
}
```

// The output will be 20

Corrected Code

```
public class Main {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 5;  
        int result = a + b * 2; // Correctly follows operator precedence  
        System.out.println(result);  
    }  
}
```

Snippet 19:

```
public class Main {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 0;  
        int result = a / b; // Division by zero  
        System.out.println(result);  
    }  
}
```

Exception in thread "main" java.lang.ArithmeticException: / by zero

Corrected Code

```
public class Main {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 0;  
        if (b != 0) {  
            int result = a / b;
```

```

        System.out.println(result);
    } else {
        System.out.println("Cannot divide by zero");
    }
}
}

```

Snippet 20:

```

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello, World")
    }
}

```

error: ';' expected

Corrected Code

```

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello, World!"); // Added missing semicolon
    }
}

```

Snippet 21:

```

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
        // Missing closing brace here
    }
}

```

error: reached end of file while parsing

Corrected Code

```

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    } // Added missing closing brace
}

```

Snippet 22:

```
public class Main {  
    public static void main(String[] args) {  
        static void displayMessage() {  
            System.out.println("Message");  
        }  
    }  
}
```

error: illegal static declaration; non-static method displayMessage() cannot be referenced from a static context

Corrected Code

```
public class Main {  
    public static void main(String[] args) {  
        displayMessage(); // Calling the method from within the main method  
    }  
  
    static void displayMessage() {  
        System.out.println("Message");  
    }  
}
```

Snippet 23:

```
public class Confusion {  
    public static void main(String[] args) {  
        int value = 2;  
        switch(value) {  
            case 1:  
                System.out.println("Value is 1");  
            case 2:  
                System.out.println("Value is 2");  
            case 3:  
                System.out.println("Value is 3");  
            default:  
                System.out.println("Default case");  
        }  
    }  
}
```

// Why does the default case print after "Value is 2"?

// The missing break statement in each case causes "fall-through" behavior, meaning the program continues executing subsequent cases even if they don't match the value.

Corrected Code

```
public class Confusion {  
    public static void main(String[] args) {  
        int value = 2;  
        switch(value) {  
            case 1:  
                System.out.println("Value is 1");  
                break;  
            case 2:  
                System.out.println("Value is 2");  
                break;  
            case 3:  
                System.out.println("Value is 3");  
                break;  
            default:  
                System.out.println("Default case");  
        }  
    }  
}
```

Snippet 24:

```
public class MissingBreakCase {  
    public static void main(String[] args) {  
        int level = 1;  
        switch(level) {  
            case 1:  
                System.out.println("Level 1");  
            case 2:  
                System.out.println("Level 2");  
            case 3:  
                System.out.println("Level 3");  
            default:  
                System.out.println("Unknown level");  
        }  
    }  
}
```

```
}  
}
```

// Why does it print all cases after "Level 1"?

// The missing break statements cause the program to execute all following cases after a match is found.

Corrected Code

```
public class MissingBreakCase {  
    public static void main(String[] args) {  
        int level = 1;  
        switch(level) {  
            case 1:  
                System.out.println("Level 1");  
                break;  
            case 2:  
                System.out.println("Level 2");  
                break;  
            case 3:  
                System.out.println("Level 3");  
                break;  
            default:  
                System.out.println("Unknown level");  
        }  
    }  
}
```

Snippet 25:

```
java  
Copy  
public class Switch {  
    public static void main(String[] args) {  
        double score = 85.0;  
        switch(score) {  
            case 100:  
                System.out.println("Perfect score!");  
                break;  
            case 85:
```

```

        System.out.println("Great job!");
        break;
    default:
        System.out.println("Keep trying!");
    }
}
}

```

error: switch expression must be of type int, char, byte, short, or an enum type

Corrected Code

```

public class Switch {
    public static void main(String[] args) {
        int score = 85;
        switch(score) {
            case 100:
                System.out.println("Perfect score!");
                break;
            case 85:
                System.out.println("Great job!");
                break;
            default:
                System.out.println("Keep trying!");
        }
    }
}

```

Snippet 26:

```

public class Switch {
    public static void main(String[] args) {
        int number = 5;
        switch(number) {
            case 5:
                System.out.println("Number is 5"); break;
            case 5:
                System.out.println("This is another case 5");
                break;
            default:
                System.out.println("This is the default case");
        }
    }
}

```

```
}
```

error: duplicate case label

Corrected Code

```
public class Switch {  
    public static void main(String[] args) {  
        int number = 5;  
        switch(number) {  
            case 5:  
                System.out.println("Number is 5");  
                break;  
            default:  
                System.out.println("This is the default case");  
        }  
    }  
}
```